CS182: Introduction to Machine Learning
Lecture 04: Optimization Primer

Lecture Quiz and Answer Copy

Scope: Optimization_Primer.pdf. Independent practice material based on the official 2026 Fall lecture.

Total: 100 marks. **Suggested time:** 120 minutes.

Name: _____ **Student ID:** _____

Section	Questions	Each	Marks
A. Misconceptions: True/False	12	1	12
B. Multiple response	12	2	24
C. Proofs and formula derivations	4	10	40
D. Applications	2	12	24
Total	30		100

Scoring. In Section B each item has exactly four options, of which one, two, or three are correct. The complete correct set earns 2 marks; a nonempty proper subset earns 1 mark; any incorrect option or no selection earns 0. T/F answers earn 1 mark only when correct. Show reasoning in Sections C and D; part marks are available for valid intermediate work. Equivalent correct methods receive credit.

Instructions. Answer in English. Use natural logarithms. Exact values and justified approximations are accepted. All vectors are columns unless stated otherwise. Use local assumptions, label conventions, and tie rules.

Answer sheet — Section A

1	2	3	4	5	6

7	8	9	10	11	12

Answer sheet — Section B

13	14	15	16	17	18

19	20	21	22	23	24

Section A. True/False 12 marks

1. The feasible set defined by convex inequalities $f_i(x) \leq 0$ and affine equalities is convex.

2. For a symmetric matrix Q, the unconstrained quadratic $f(x) = \frac{1}{2}x^TQx + c^Tx$ is convex if and only if $Q \succeq 0$.

3. The Lagrange dual function is concave even when the primal objective or inequality constraints are nonconvex.

4. If a constraint is active at a KKT point, its multiplier must be strictly positive.

5. If a primal feasible point and a dual feasible point have the same finite objective value, they certify primal and dual optimality.

6. A differentiable function is convex whenever its gradient vanishes at at least one point.

7. Every local minimizer of a convex function over a convex feasible set is a global minimizer.

8. For differentiable convex objectives, every positive constant gradient-descent step size guarantees convergence from every starting point.

9. Equality multipliers may have either sign because the corresponding constraint values are zero at every primal feasible point.

10. Every differentiable convex problem with an attained optimum has an optimal finite multiplier vector.

11. At a KKT point, a strictly positive inequality multiplier implies that the corresponding inequality is active.

12. For a minimization problem with constraints $f_i(x) \leq 0$, every real multiplier vector provides a valid dual lower bound.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

13. Let f be differentiable and convex on $\mathbb{R}^n$. Select every conclusion that follows.

(A) The Hessian must be positive definite wherever it exists.

(B) If $\nabla f(x) = 0$, then x is a global unconstrained minimizer.

(C) $f(y) \geq f(x) + \nabla f(x)^T(y - x)$ for all x, y.

(D) Every global minimizer is unique.

14. For a general primal minimization problem, select every statement justified without assuming convexity or strong duality.

(A) The dual function is concave on its effective domain.

(B) A zero gradient at a primal feasible point guarantees the global constrained optimum.

(C) Any primal feasible x and dual feasible multipliers satisfy $g(\lambda, \nu) \leq f_0(x)$.

(D) If a differentiable Lagrangian has zero gradient at x, that point necessarily realizes its infimum.

15. Consider minimizing $(x - 3)^2$ subject to $x \leq 1$, with $L(x, \lambda) = (x - 3)^2 + \lambda(x - 1)$. Select every correct calculation.

(A) $g(\lambda) = 2\lambda - \lambda^2/4$.

(B) The unconstrained minimizer of $L(\cdot, \lambda)$ is $x = 3 - \lambda/2$.

(C) The dual optimum is attained at $\lambda = 4$, with value 4.

(D) $\lambda = -2$ gives a valid dual certificate because the Lagrangian infimum is finite.

16. Which formulations satisfy the definition of a convex optimization problem? All variables shown are real.

(A) Minimize $-x^2$ subject to $-1 \leq x \leq 1$.

(B) Minimize x subject to $x^2 = 1$.

(C) Minimize $x^2 + y^2$ subject to $x + y = 1$ and $x \geq 0$.

(D) Minimize $3x - y$ subject to $x^2 + y^2 \leq 4$.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

17. Let $G = \begin{pmatrix} 2 & 6 \\ 0 & 4 \end{pmatrix}$ and $f(x) = \frac{1}{2}x^T G x$ on $\mathbb{R}^2$. Select every correct statement.

(A) $\nabla f(x) = (2x_1 + 3x_2,\, 3x_1 + 4x_2)^T$.

(B) f is convex because the diagonal entries of G are positive.

(C) The Hessian equals the nonsymmetric matrix G.

(D) The stationary point $x = 0$ is a saddle point.

18. Select every correct statement about stopping gradient descent or checking optimality.

(A) The norm $\|\nabla f(x)\|$ can be used as a scalar stationarity residual.

(B) A small gradient norm alone gives a universal numerical bound on objective error for every differentiable convex function.

(C) A zero gradient certifies a global unconstrained minimizer for a differentiable convex objective.

(D) For a constrained minimizer on the boundary, the objective gradient must itself be zero.

19. For minimizing $f_0(x)$ subject to $f_i(x) \le 0$ and $h_j(x) = 0$, select every correct Lagrangian or dual convention.

(A) The dual function is obtained by minimizing the Lagrangian only over primal feasible points.

(B) The dual problem maximizes $g(\lambda, \nu)$ over $\lambda \ge 0$.

(C) If the infimum of the Lagrangian is $-\infty$, that multiplier vector is dual feasible under the lecture definition.

(D) $L = f_0 + \sum_i \lambda_i f_i + \sum_j \nu_j h_j$, with $\lambda \ge 0$ and unrestricted ν.

20. The primal linear program is $\min_x c^T x$ subject to $Ax \le b$, where $A \in \mathbb{R}^{m \times n}$ and x is unrestricted. Select every correct statement.

(A) The dual has n variables and m equality constraints.

(B) Its dual maximizes $-b^T \lambda$ subject to $A^T \lambda + c = 0$ and $\lambda \ge 0$.

(C) If $A^T \lambda + c \neq 0$, the dual function equals $-\infty$.

(D) The nonnegativity restriction on λ can be dropped because x is unrestricted.

Section B. Multiple Response 24 marks

Select all correct options. Each question has 1–3 correct options.

21. For $f(x) = \frac{1}{2}(x_1^2 + 4x_2^2)$, gradient descent uses a constant step size $\eta = 0.6$. Select every correct conclusion.

(A) The iterates converge to zero from every starting point because the objective is strictly convex.

(B) The update is $x_{k+1} = (0.4x_{1,k}, -1.4x_{2,k})^T$.

(C) At any starting point with $x_{2,0} \neq 0$, the second coordinate changes sign each iteration and grows in magnitude.

(D) If $x_{2,0} = 0$, the iterates converge to zero.

22. Let f be convex and let X be a finitely supported random vector in its convex domain. Select every valid use of convexity.

(A) The pointwise maximum of two convex functions is convex.

(B) The pointwise minimum of any two convex functions is convex.

(C) Jensen's inequality requires independent coordinates of X.

(D) $f(E[X]) \geq E[f(X)]$ for every such convex function and random vector.

23. Minimize x^2 subject to $-x \leq 0$ and $-2x \leq 0$. Select every correct statement at $x^* = 0$.

(A) $(x^*, \lambda_1, \lambda_2) = (0, 1, 0)$ satisfies all KKT conditions.

(B) Both inequalities are active.

(C) The stationarity equation is $2x - \lambda_1 - \lambda_2 = 0$.

(D) Both multipliers must be strictly positive.

24. An algorithm has a primal feasible point with objective 11.2 and dual feasible multipliers with dual value 10.9. Select every justified conclusion.

(A) The optimal value must equal the midpoint 11.05.

(B) The distance from the primal point to every minimizer is at most 0.3.

(C) The primal point is at most 0.3 above the optimal objective value.

(D) The optimal primal value belongs to $[10.9, 11.2]$.

Section C. Proofs and Formula Derivations

25. Proving the first-order characterization of convexity **[10 marks]**

(a) Let f be differentiable on an open convex domain. Starting with the defining convexity inequality, prove that convexity implies $f(y) \geq f(x) + \nabla f(x)^T(y - x)$ for any x, y in the domain. **[5 marks]**

(b) Conversely, assume the first-order inequality holds for every pair x, y. Prove the defining convexity inequality using $z = (1 - t)x + ty$, $0 < t < 1$, as the expansion point. **[5 marks]**

Student response

(a)

(b)

Section C. Proofs and Formula Derivations

26. Dual concavity, weak duality, and complementary slackness **[10 marks]**

(a) For $L(x, \lambda, \nu) = f_0(x) + \sum_i \lambda_i f_i(x) + \sum_j \nu_j h_j(x)$, prove directly that $g(\lambda, \nu) = \inf_x L(x, \lambda, \nu)$ is concave, without imposing convexity on the primal functions. Work with multiplier pairs having finite dual values. **[4 marks]**

(b) For primal feasible x and $\lambda \geq 0$, derive $g(\lambda, \nu) \leq f_0(x)$ and hence weak duality. Explain the different restrictions on inequality and equality multipliers. **[3 marks]**

(c) Suppose feasible x^*, λ^*, ν^* have finite values and $f_0(x^*) = g(\lambda^*, \nu^*)$. Derive complementary slackness term by term and show that x^* minimizes the Lagrangian. **[3 marks]**

Student response

(a)

(b)

(c)

Section C. Proofs and Formula Derivations

27. Why convex KKT conditions are sufficient **[10 marks]**

(a) Assume f_0, f_i are differentiable convex functions on $\mathbb{R}^n$, and h_j are affine. Suppose (x^*, λ^*, ν^*) satisfies primal feasibility, $\lambda^* \geq 0$, complementary slackness, and stationarity. Prove that x^* globally minimizes $L(\cdot, \lambda^*, \nu^*)$. **[4 marks]**

(b) Use this Lagrangian minimum to prove primal and dual optimality and zero duality gap. State whether a constraint qualification is needed for this sufficiency implication once the displayed KKT point is given. **[3 marks]**

(c) If f_0 is strictly convex, prove that the primal minimizer is unique. Identify the role of the convexity of the feasible set. **[3 marks]**

Student response

(a)

(b)

(c)

Section C. Proofs and Formula Derivations

28. Deriving a quadratic gradient-descent step-size bound **[10 marks]**

(a) Let $f(x) = \frac{1}{2}x^T Qx + c^T x + r$, where Q is symmetric. Derive its gradient and Hessian by expanding $f(x+h)$. State the condition for convexity on $\mathbb{R}^n$. **[3 marks]**

(b) Assume $0 \preceq Q \preceq LI$ with $L > 0$, let $g = \nabla f(x)$, and define $x^+ = x - \eta g$. Derive an upper bound on $f(x^+) - f(x)$, and give a range of η guaranteeing strict decrease whenever $g \neq 0$. **[4 marks]**

(c) Now assume $Q \succ 0$. Prove convergence of constant-step gradient descent for $0 < \eta < 2/\lambda_{\max}(Q)$ by diagonalizing Q. Give a one-dimensional counterexample at the upper endpoint. **[3 marks]**

Student response

(a)

(b)

(c)

Section D. Applications

29. Capacity planning with an active resource cap **[12 marks]**

(a) A planning model chooses nonnegative production amounts x, y with total $x + y = 4$, subject to the capacity cap $x \leq 2.5$. It minimizes $f(x, y) = (x - 5)^2 + 2(y - 2)^2$. Eliminate the equality constraint, determine the feasible interval, and find the unique minimizer and objective value. **[4 marks]**

(b) Write the Lagrangian using multipliers $\alpha, \beta, \lambda \geq 0$ for $-x, -y, x - 2.5 \leq 0$, respectively, and unrestricted ν for $x + y - 4 = 0$. Find multipliers at the proposed optimum and give a dual certificate by evaluating the Lagrangian infimum. **[4 marks]**

(c) Before convergence, a solver reports the feasible plan $(x, y) = (2, 2)$ and multipliers $(\alpha, \beta, \lambda, \nu) = (0, 0, 2, 2)$. Compute their dual value, a guaranteed interval for p^*, and an upper bound on the plan's objective error. **[4 marks]**

Student response

(a)

(b)

(c)

Section D. Applications

30. Choosing a stable learning rate for a quadratic calibration loss [12 marks]

(a) A calibration model minimizes $f(u, v) = \frac{1}{2}(2u^2 + 8v^2) - 4u + 8v$ on $\mathbb{R}^2$. Compute the gradient, Hessian, unique optimizer, and minimum value. Explain why a stationary point is sufficient here.**[4 marks]**

(b) From $(u_0, v_0) = (0, 0)$, calculate one gradient-descent step and its objective for $\eta = 0.2$ and for $\eta = 0.3$. Compare each objective to the initial value and to the optimum. **[4 marks]**

(c) Derive the constant-step interval guaranteeing convergence from every start. At the boundary step $\eta = 0.25$, determine the behavior from $(0, 0)$ and explain why a zero change in objective would not certify convergence. **[4 marks]**

Student response

(a)

(b)

(c)